

CBCS SCHEME

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BBOC407

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Biology for Engineers (CSE)

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Discuss the various components of Eukaryotic cells.	10	L3	CO1
	b.	Identify the applications of stem cells.	5	L2	CO1
	c.	Explain the functions of vitamins.	5	L2	CO1
OR					
Q.2	a.	Compare Prokaryotic and Eukaryotic cells.	10	L3	CO1
	b.	Explain the properties of Carbohydrates.	5	L2	CO1
	c.	Explain the functions of Lipids.	5	L2	CO1
Module – 2					
Q.3	a.	Highlighting the properties of cellulose, justify cellulose as an effective water filter.	10	L3	CO1
	b.	Explain the working and development of DNA vaccines by taking suitable example.	10	L2	CO1
OR					
Q.4	a.	What are Bioplastics? Justify the use of PHA as Bioplastic mentioning its properties and applications.	10	L3	CO1
	b.	Discuss the following : (i) Meat analogs of protein. (ii) Lipids as cleaning agents.	10	L2	CO1
Module – 3					
Q.5	a.	What is Electro Encephalogram (EEG)? Discuss the types of Brain activity detected with EEG. Write any three applications.	10	L3	CO2
	b.	What are Pace Makers? Explain basic design and construction of Pace Makers.	10	L2	CO2
OR					
Q.6	a.	Justify Lungs as purification system.	10	L3	CO2
	b.	Explain architecture of Rod and Core cells with suitable diagram.	10	L2	CO2
Module – 4					
Q.7	a.	What is ultrasonography? Explain the uses and working principle.	10	L2	CO3
	b.	What is lotus leaf effect? Explain the mechanism and applications of super Hydrophobic effect.	10	L2	CO3
OR					
Q.8	a.	The structure and design of Kingfisher beak lead to the design of Bullet trains. Explain.	10	L2	CO3
	b.	Explain the working and applications of Bionic Leaf Technology.	10	L2	CO3

Module – 5					
Q.9	a.	Explain the use of Electrical tongue in food science.	10	L2	CO4
	b.	Explain the advantages and limitations of Artificial Intelligence for disease diagnosis.	10	L2	CO4
OR					
Q.10	a.	Explain Bioengineering solutions for muscular dystrophy and Osteoporosis.	10	L2	CO4
	b.	Explain most commonly used Bioprinting Techniques.	10	L2	CO4



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Belagavi, Karnataka - 590 018.

Scheme & Solutions

Signature of Scrutinizer

Subject Title : **BIOLOGY FOR ENGINEERS** Subject Code : **BB0C407**

Question Number	Solution	Marks Allocated
1. a)	components of Eukaryotic cells Cell membrane, cytoplasm, organelles, Nucleus, Nucleolus, Mitochondria, Ribosome, Endoplasmic Reticulum, Golgi Bodies, Vacuole, Vesicles etc. Explanation of any 10, 1 mark each.	10x1=10.
b)	Applications of Stem cells - 1) HSC Transplantation 2) Placental stem therapy. 3) Artificial organ Engineering. 4) Anti-aging effects 5) wound healing Explanation of any five applications	5
c)	Functions of Vitamins - Vitamin A - Vision, Skin health Vitamin B - RBC formation, Nerve function Vitamin C - Boosts Immune system, antioxidant Vitamin D - Bone health, Calcium absorption Vitamin E - Protecting cells from damage Vitamin K - Blood clotting, Bone health etc. Any 5 - 1 mark each	5

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2) a)	<table border="1"> <thead> <tr> <th>Parameter</th><th>Prokaryotic</th><th>Eukaryotic</th></tr> </thead> <tbody> <tr> <td>Nucleus</td><td>Absent</td><td>Present</td></tr> <tr> <td>cell size</td><td>smaller</td><td>larger</td></tr> <tr> <td>cell structure</td><td>unicellular</td><td>Most Multicellular</td></tr> <tr> <td>complexity</td><td>Simpler</td><td>complex</td></tr> <tr> <td>DNA</td><td>circular</td><td>linear</td></tr> <tr> <td>Mitochondria</td><td>Absent</td><td>Present</td></tr> <tr> <td>Golgi Apparatus</td><td>Ab</td><td>Present</td></tr> <tr> <td>Reproduction</td><td>Asexual</td><td>Sexual</td></tr> <tr> <td>cell wall</td><td>Present</td><td>absent</td></tr> <tr> <td>Example</td><td>Bacteria</td><td>Fungi, animal, Plant</td></tr> </tbody> </table>	Parameter	Prokaryotic	Eukaryotic	Nucleus	Absent	Present	cell size	smaller	larger	cell structure	unicellular	Most Multicellular	complexity	Simpler	complex	DNA	circular	linear	Mitochondria	Absent	Present	Golgi Apparatus	Ab	Present	Reproduction	Asexual	Sexual	cell wall	Present	absent	Example	Bacteria	Fungi, animal, Plant	1x10
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b)	Properties - Physical - Stereoisomerism optical activity Anomerism Chemical - Osazone formation oxidation Reduction to alcohols.	5																																	
c)	Functions of Lipids Energy storage, Insulation, cell membrane structure, Hormone synthesis, Transport with explanation (mark each)	1x5																																	
3) a)	Properties - High Porosity Biodegradability, cost effective, Good mechanical strength, chemical resistance Explanation of working	05 05																																	
b)	DNA vaccine - is a type of vaccine that uses a piece of viral or bacterial DNA to stimulate an immune response against pathogen	02																																	

Question Number	Solution	Marks Allocated
	working -	3
	DNA vaccine for Rabies, Importance -	5
4) a)	Bio plastics - Biodegradable and Biocompatible	02
	Properties of BHA -	04
	Applications -	04
b) (i)	Meat analogs & Protein	
	- Meat analogs - meat substitutes are plant based foods designed to mimic the taste, texture and appearance of meat.	02
	Explanations with examples - Tofu, Tempeh, Seitan, Veggie Burgers, Plant based Sausage.	03
(ii)	Lipids as cleaning agents -	
	Explanation with examples -	03
	Advantages and limitations (2 each)	02
5) a)	EEG - Non invasive method for measuring and recording of the electrical activity of the brain.	02
	Types of Brain activity - Delta waves (0.5-4 Hz)	
	Theta wave (4-8 Hz), Alpha wave (8-12 Hz)	05
	Beta wave (12-30 Hz), Gamma wave (30-100 Hz) with explanation.	
	Any 3 applications	03
5) b)	A pacemaker is a small device that surgically implanted in the chest to regulate the heart beat.	02
	Type - Design with explanation, construction	04
		04

Question Number	Solution	Marks Allocated
6) a)	The lung purifies air by removing harmful substances and adding oxygen to the blood stream, process - Filtration, Moisturization, Gas Exchange with explanation	02. 03 05
b)	<u>Rod cells</u> - are photoreceptor cells in the retina of the eye that are responsible for detecting light and transmitting signals to the brain for perception of vision especially in low light condition Explanation with diagram	02 03
	<u>Cone cells</u> - photoreceptor cells in the retina that are responsible for color vision and visual acuity. Explanation with diagram	02 03
7) a)	Ultrasonography is a medical imaging technique that uses high freq. sound waves to produce images of the internal organs and tissues of the body. Uses - Gynecology, Abdominal Imaging, Musculoskeletal Imaging, Vascular Imaging, Eye and Neck Imaging, Emergency medicine. working	02 04. 04.
b)	The ability of lotus leaves to repel water and self clean through their unique surface structure. Mechanism of Super Hydrophobic Effect Applications - Electronics Industry, Automobile Ind, Aerospace Industry, with explanation	02 04. 04

Question Number	Solution	Marks Allocated
8) a)	Physics behind Kingfisher - Beak streamlining, Surface Tension, Minimizing splash with explanation Technological Importance with explanation	05 05
b)	Bionic leaf - which aims to mimic the process of photosynthesis in plants. consists of photovoltaic cell - captures sunlight and convert it into electrical energy. Applications - Renewable Energy production, CO ₂ Reduction, Agriculture and food production, etc. with explanation (any 5)	05 05
9) a)	Electrical tongue is a used to analyze the taste and flavor of food and beverage. Technology and working Advantages - Non Invasive, High throughput, objective analysis, cost effective.	02 04 04
b)	Advantages - Image analysis, Data analysis, Diagnose, Medicine, clinical decision support with explanation Limitation - any 5 with explanation	05 05
10) a)	Bio Engineering soln for muscular dystrophy any 5 - Gene Therapy, Stem cell therapy, Tissue engineering, Exoskeleton technology Bio Engineering soln for osteoporosis Tissue Eng, Stem cell therapy, Biomaterials, Gene therapy	05 05
b)	Bioprinting - working explanation Advantages & limitation.	05 05